

Vector Databases Explained: Powering AI with Open Source Data Storage

What do vector databases do? Why did they show up when they did? Which open source ones are worth your time? How do you use them? Here are some answers...



OPEN SOURCE DATA STORAGE

and explained the answer to anyone properly. I had been using vector databases for over a year, but mostly through libraries and convenience tools, without ever really articulating why they exist as a separate category. So we grabbed some coffee and I tried to explain the choice.

This article is based on that conversation because I think a lot of developers are in his position right now. They are building AI features and keep hearing about vector databases. They sort of know what embeddings are. But the whole stack feels like a black box, and the open source options are multiplying faster than anyone can keep up with.

The problem vector databases solve

Traditional databases are built for exact matches. Give me the user with ID 7421. Give me all orders placed between Tuesday and Thursday. Give me products where the price is greater than five hundred. These are all commands with crisp, well-defined answers, and SQL is brilliant at them.

Then AI showed up and broke the model. Suddenly the questions people wanted to ask were fuzzier. Find me documents similar in meaning to this paragraph. Suggest products that look like this one. Show me support tickets that sound like the one this customer just opened. These aren't exact match

A couple of months ago, a junior developer on a team I was helping asked me an interesting question. He had been working on a small support chatbot to answer customers' questions using their internal documentation. He'd hooked it up to an LLM, written some decent prompts,

and the bot worked reasonably well. But he wanted to know why we were using something called Qdrant for storage instead of just MySQL, since the latter was already in their stack and everyone was familiar with it.

It was a fair question. And I realised I'd never actually sat down